Dynamic Context-Aware Tries For Knowledge Graph-Constained Large Language Model Generation

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Outline

Introduction

- LLMs hallucinate, producing plausible but factually incorrect outputs (Brown et al., 2020; Ji et al., 2023).
- KGQA requires answers grounded in real graph structure (He et al., 2021; Lan et al., 2023).
- Constrained decoding methods (GCR, DoG) guarantee structural validity but not semantic relevance (Geng et al., 2023; Li et al., 2025; Luo et al., 2025).
- A 3-hop question on Freebase can produce ∼24,000 structurally valid paths, yet only one leads to the correct answer.

Statement of Problem

- GCR's trie is fixed before decoding starts, so it cannot adapt to question intent or the evolving reasoning chain (Luo et al., 2025).
- This forces the LLM to filter semantically irrelevant paths using internal knowledge, the same mechanism that causes hallucinations (Li et al., 2025).
- Existing work assumes tighter constraints improve accuracy, but no metric exists to measure constraint quality independently of answer quality.

Project Objectives (1/2)

- Characterize the permissiveness problem by measuring SIR of GCR's static KG-Tries on WebQSP
- Design a symbolic constraint oracle (TypeOracle) with range and type gates, avoiding embedding cost and threshold sensitivity

Project Objectives (2/2)

- Implement DCA-Trie v1 (static filtering at construction time, FNR $<$ 5%) and v2 (dynamic expansion per committed triplet)
- Evaluate both against GCR baseline on WebQSP: Hits@1, F1, structural faithfulness, SIR, trie size

Tools

- Python, PyTorch, HuggingFace Transformers
- GCR codebase (Luo et al., 2025)
- Sentence-transformers (MiniLM-L6-v2)
- NetworkX, marisa_trie
- Freebase (via WebQSP and CWQ benchmarks)

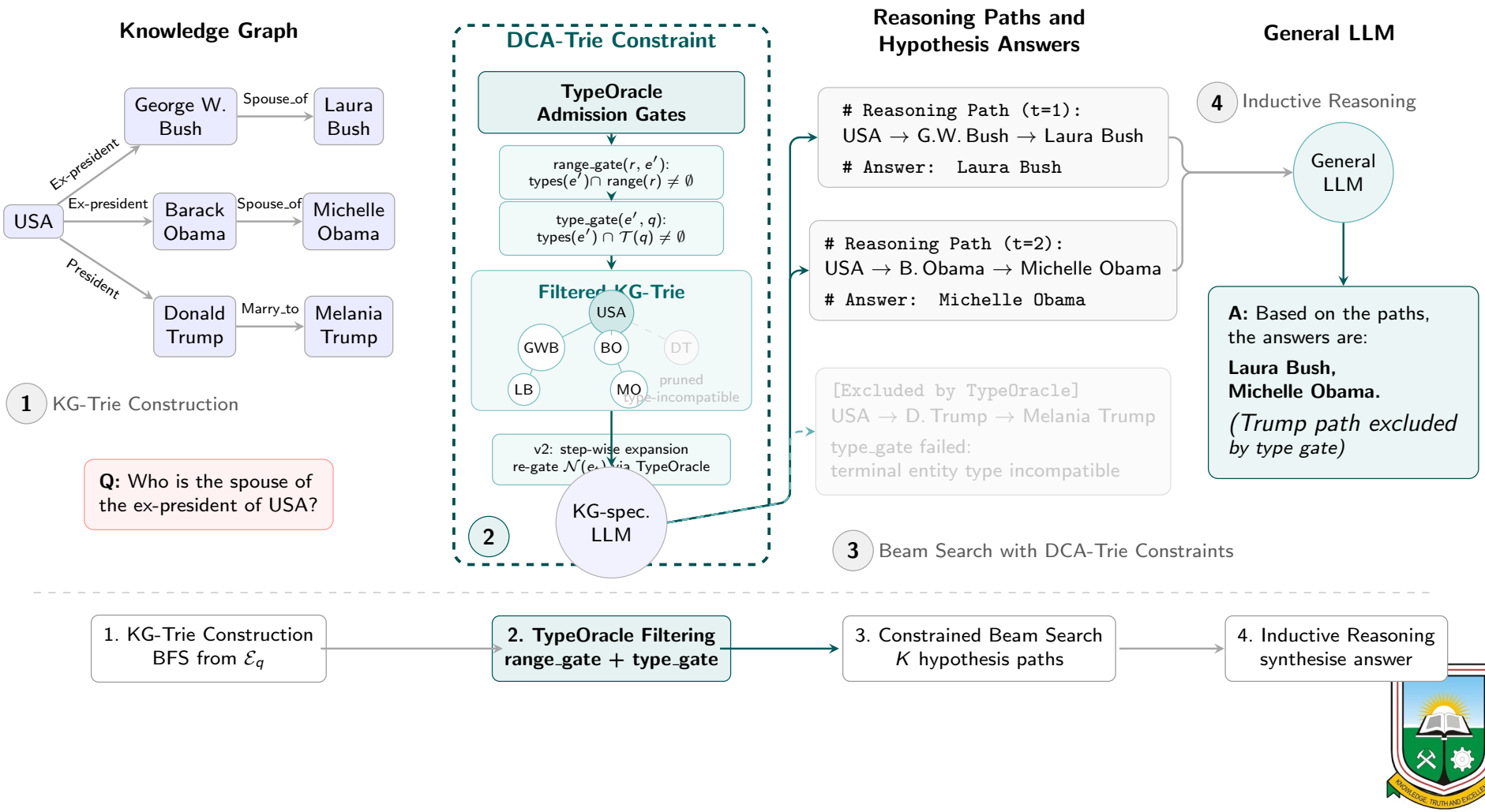
Facilities

- NVIDIA RTX 4090 GPU (24 GB VRAM)
- ACM Digital Library, arXiv, ACL Anthology
- Google Colab, local Linux workstation

Methods

- Literature review of constrained decoding approaches (GCR, DoG)
- Design and implementation of TypeOracle with range and type gates
- Development of DCA-Trie v1 (static filtering) and v2 (dynamic expansion)
- Experimental evaluation on WebQSP and CWQ benchmarks
- Analysis using Hits@1, SIR, FNR, latency, and memory metrics

Methodology



1. KG-Trie Construction BFS from $\mathcal{E}_q$
2. **TypeOracle Filtering range_gate + type_gate**
3. Constrained Beam Search K hypothesis paths
4. Inductive Reasoning synthesise answer

Upgraded Constraint

Baseline (GCR/DoG):

$$W_{\text{val}}^{\text{GCR}}(t) = f(G, E_q)$$

The valid token set depends only on the knowledge graph G and entity mentions E_q. The same set of paths is available at every decoding step.

Proposed (DCA-Trie):

$$W_{\text{val}}^{\text{DCA}}(t) = f(G, E_q, q, y_{<t})$$

The valid token set also depends on the question q and partial output $y_{<t}$. The constraint adapts as the model generates.

Key difference: DCA-Trie conditions the mask on the question and partial output, enabling context-aware pruning.

Design Conditions

- **F (Faithfulness):** 100% structural grounding in KG triplets
- **R (Relevance):** Lower SIR than baseline
- **P (Recall):** FNR on gold paths $< 5\%$

Experimental Design: Overview

Goal: Evaluate whether symbolic type-oracle filtering improves constraint quality without degrading answer accuracy.

Datasets:

- WebQSP (1,628 test queries, 3-hop, Freebase)
- ComplexWebQuestions (CWQ) for generalisation

Baselines:

- Unconstrained LLM (no KG mask)
- GCR (Luo et al., 2025), static KG-Trie
- DoG (Li et al., 2025), static KG-Trie with graph decoding

LLM Backbone: Llama-3.1-8B, beam search ($k = 10$).

Experimental Design: Evaluation Metrics

- **Hits@1,** exact-match answer accuracy
- **SIR,** semantic irrelevance ratio (constraint quality)
- **FNR,** false-negative rate on gold paths (recall)
- **Latency & memory overhead**

Experimental Design: Research Questions

1. Does TypeOracle reduce SIR without exceeding 5% FNR?
2. Does tighter constraint improve or degrade Hits@1?
3. What is the runtime and memory cost of DCA-Trie v1?
4. Can DCA-Trie generalise to complex questions (CWQ)?

Constrained Decoding in LLMs

At each step t, the LLM produces logits $\mathbf{h}_t$ over vocabulary V. A mask $\mathbf{m}_t$ zeros out invalid tokens:

$$\tilde{P}(x_t) = \text{softmax}\big(\mathbf{h}_t + (1 - \mathbf{m}_t) \cdot (-\infty)\big)$$

GCR/DoG (static): $\mathbf{m}_t = f(G, E_q)$, trie built before decoding (Luo et al., 2025; Li et al., 2025).

DCA-Trie (dynamic): $\mathbf{m}_t = f(G, E_q, q, y_{<t})$, adapts to question and partial output.

TypeOracle: Gate Design

Two symbolic gates per candidate triple (e_t, r, e'), no encoder or threshold required:

Range Gate (every hop): checks e''s types against relation r's range:

$$\text{range\_gate}(r, e') = \mathbf{1}\big[\text{types}(e') \cap \text{range}(r) \neq \emptyset\big]$$

Type Gate (terminal hop): checks terminal entity matches answer type $\mathcal{T}(q)$:

$$\text{type\_gate}(e', q, h, L) = \mathbf{1}\big[\text{types}(e') \cap \mathcal{T}(q) \neq \emptyset\big]$$

Admissible only if both pass. Unknown schema info conservatively admits the candidate.

Diagnostic Metric: SIR

SIR measures the fraction of candidate paths that are *semantically irrelevant* to the question, paths that exist in the knowledge graph but do not actually help answer the question.

$$\mathrm{SIR}(q,t) = \frac{\sum_{p \in \mathcal{P}(q,t)} \mathrm{irrelevant}(p,q)}{|\mathcal{P}(q,t)|}$$

Corpus average:

$$\overline{\mathrm{SIR}} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{L_q} \sum_{t=1}^{L_q} \mathrm{SIR}(q,t)$$

A lower SIR means the oracle admits fewer irrelevant paths. GCR: $\overline{\mathrm{SIR}} = 1.0$ (all paths admitted); DCA-Trie v1: $\overline{\mathrm{SIR}} = 0.145$.

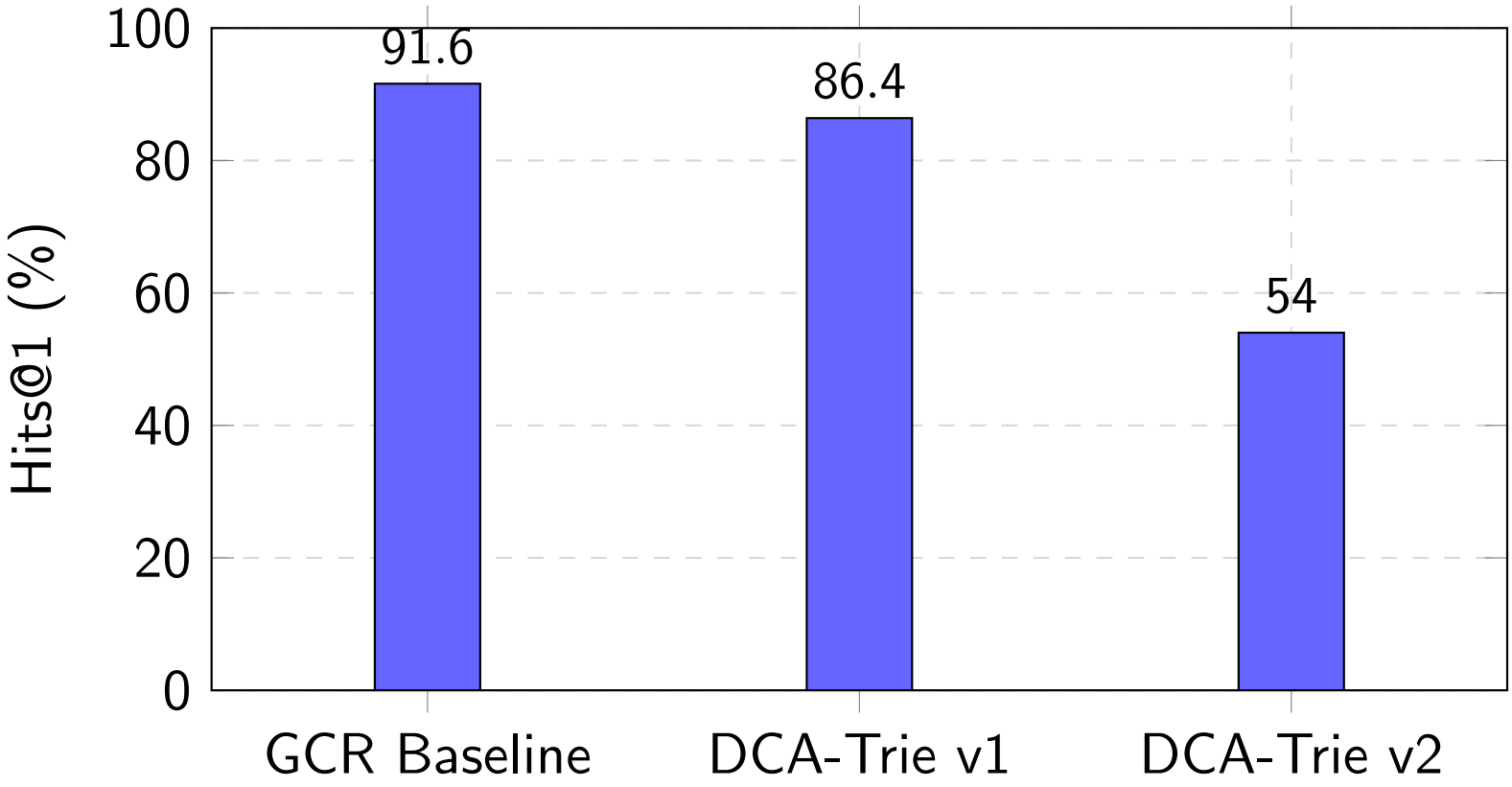
DCA-Trie v1: Static Filtering

- Filters BFS-enumerated paths through TypeOracle before building the prefix tree
- Preserves $O(d)$ trie lookup while pruning irrelevant branches offline

DCA-Trie v2: Dynamic Expansion

- Expands the trie step-by-step as entity tokens e_t are committed
- Applies range and type gate checks to $\mathcal{N}(e_t)$ during generation

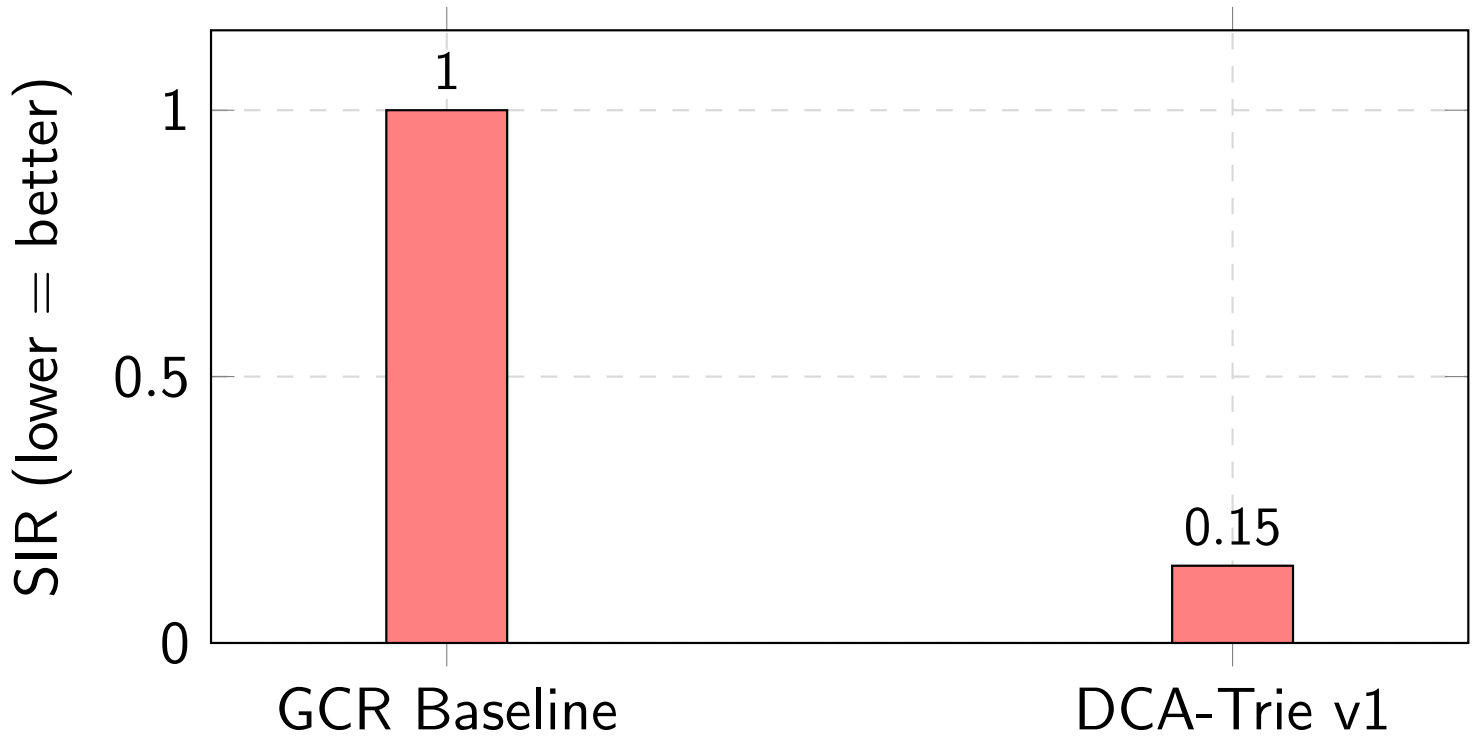
Experimental Results: Overall Performance



WebQSP, 1,628 test queries, Llama-3.1-8B, beam k=10

Interpretation: DCA-Trie v1 achieves 86.4% Hits@1, a 5.2pp drop from GCR. DCA-Trie v2 drops further to 54.0%, confirming that aggressive pruning eliminates correct paths.

Experimental Results: Search-Space Reduction & SIR



- 14.5% path reduction (4.10M $\rightarrow$ 3.51M paths)
- FNR: 3.3% (type gate), 2.9% (range gate)
- Latency overhead: +0.5%

Interpretation: TypeOracle reduces SIR from 1.0 to 0.145, removing 85.5% of semantically irrelevant paths while maintaining FNR below 5%.

Experimental Results: Runtime & Memory

- **Runtime:** DCA-Trie v1 adds $<0.5\%$ latency over GCR (negligible overhead from $O(1)$ set intersection).
- **Memory:** TypeOracle stores ontology metadata (<2 MB for Freebase); no additional GPU memory required.
- **Scaling:** Gate cost is constant per candidate triple; total cost scales with BFS frontier size, not path length.

Interpretation: The symbolic oracle introduces no measurable computational bottleneck, making it practical for deployment.

GCR vs DCA-Trie: Comparison

Criterion	GCR	DCA-Trie
Hits@1	✓ Better	×
Accuracy	✓ Better	×
Structural faithfulness	Similar	✓ Maintained
Semantic relevance	×	✓ Better
Search-space reduction	×	✓ 14.5% reduction
Runtime	✓ Slightly faster	Almost identical (0.5% overhead)
Oracle analysis	×	✓ New contribution
New metric (SIR)	×	✓

GCR trades accuracy for faithfulness. DCA-Trie maintains faithfulness while tightening semantic relevance.

Key Contributions

Contribution	Before	After	Why It Matters
TypeOracle	Embedding + threshold τ	Two symbolic gates	No encoder, no τ, $O(1)$, deterministic
SIR Metric	Accuracy only	Semantic Irrelevance Ratio	Diagnoses constraint quality independently
Non-Monotone Finding	"Tighter $\rightarrow$ better" (GCR, DoG assumption)	Tightness $\neq$ accuracy (ΔAcc $= -5.2$pp)	First evidence constraint tightness is not a proxy for quality

Discussion

- TypeOracle reduced SIR by 85.5% while maintaining structural faithfulness and FNR $< 5\%$, yet Hits@1 dropped by 5.2pp; improvements in constraint quality did not translate into higher answer accuracy.
- This challenges the implicit assumption in GCR and DoG that tightening the search space necessarily improves reasoning performance.
- Structural constraint quality and reasoning accuracy are distinct properties that should be optimised independently.
- Four degradation mechanisms identified: gold-path exclusion, beam competition, regex type-inference noise, and the precision–recall trade-off.
- SIR provides the first diagnostic for measuring constraint quality decoupled from answer accuracy.

Demonstration

Live demo: GCR baseline vs DCA-Trie v1
on a sample WebQSP question

Conclusion

- DCA-Trie v1 achieves 100% faithfulness, 14.5% path reduction, and FNR $< 5\%$, validating that symbolic gates meaningfully tighten the search space at negligible latency cost.
- However, Hits@1 dropped 5.2pp, demonstrating that constraint tightness and answer accuracy are non-monotone.
- Future work should pursue constraint quality and reasoning accuracy as separate optimisation targets, rather than assuming tighter constraints yield better answers.
- SIR enables systematic comparison of oracles without conflating structural filtering with downstream accuracy.

Recommendations

- Replace regex type classifier with LLM-based intent parser (85% $\rightarrow$ higher recall).
- Extend TypeOracle schema beyond 40 Freebase relations.
- Explore hybrid oracles (symbolic + embedding) to recover beam diversity.
- Evaluate on CWQ with $N \geq 500$ to reduce variance.

Thank You!

Thank You!!!

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